Paolo Stefano Segre

Curriculum Vitae

University of Wisconsin, Green Bay
2420 Nicolet Dr, Green Bay, WI 54311
Website: segre-ecophysiology-lab.github.io
E-mail: segrep@uwgb.edu

Education

2015 • Ph.D. Zoology, University of British Columbia, Advisor: Douglas Altshuler

2006 • M.Sc. Organismal Biology, University of Montana; Advisor: Ken Dial

2003 • B.Sc. General Biology, University of Illinois, Urbana-Champaign

Professional Appointments

2023 - Present • Assistant Professor of Biology, University of Wisconsin, Green Bay

2020 - 2023 • Lecturer, California State University, Chico

2019 - 2023 • Researcher, Stanford University

2015 - 2019 • Postdoctoral Researcher, Stanford University

Manuscripts

* denotes shared first-authorship
† denotes undergraduate co-author

2026 • Kirkham, J; Davies, KTA; Foley, HJ; Frith, R; Segre PS; Zadra, C; and SME Fortune. **Open wide: Identifying North Atlantic right whale feeding behaviour using camera-validated kinematic data and machine learning.** *PLoS One.*

2026 • den Haan, MF; Segre, PS; Kilabuk, E; Agnakak, M; Dialla, F; and SM Fortune. **Bowhead whale foraging dives are defined by speed and body orientation.** *PLoS One.*

2026 • Fish, FE; Sebo, SP; Gough, WT; Segre, PS; Immekus, R; Potvin, J; Moored, KW; and JA Goldbogen. **Delta Wing-Shaped Upper Jaw as a control surface to stabilize rorqual whales during lunge-feeding.** *Marine Mammal Science.*

2025 • Nemeth, C; Gough, WT; Segre, PS; Fish, FE; Szabo, A; Fassmann, WN; Thomson, SL; Aswegen, MV; Burrows, JA. Chenoweth, EM et al. **The key to bubble-net feeding: how humpback whale morphology functionally differs from other baleen whales.** *Journal of Experimental Biology.*

2025 • di Clemente, J; Segre, P; Garagouni, M; Winiewska, M; Kahane-Rapport, S; Gough, WT; Wahlberg, M; Goldbogen, JA; Penry, GS; Nabe-Nielsen, J; and EF Briefer. **How deep is their lunch: Inshore South African Bryde's whales shift attack mode at different depths.** *Marine Ecology Progress Series.*

2025 • di Clemente, J; Briefer, EF; Nabe-Nielsen, J; Segre, PS; Penry, GS; and M Garagouni. **"Why keep your mouth shut?" Inshore South African Bryde's whales abort lunges within feeding dives.** *Ethology, Ecology, & Evolution.*

2025 • Segre, PS. **A practical guide for improving confidence in data collected with consumer-level drones.** *Marine Mammal Science.*

2025 • Segre, PS; Dakin, R; and DL Altshuler. **Turning trade-offs: hummingbird power reserves are used to decrease turning radius or increase turning velocity.** *Journal of Zoology.*

2024 • Ayoub, A; Segre, PS; Hickmott, L; Warren, JD; Chiang, G; Bahamonde, P; and A Bocconcelli. **Feeding behavior of blue whales (*Balaenoptera musculus*) during austral summer in northern Chilean Patagonia.** *Ethology, Ecology, & Evolution.*

2024 • Fish, FE; Nicastro, AJ; Cardenas, KL; Segre, PS; Gough, WT; Kahane-Rapport, SR; Leger, JS; and JA Goldbogen. **Spin-leap performance by cetaceans is influenced by moment of inertia.** *Journal of Experimental Biology.*

2024 • Ryan, C; Martins, MCI; Healy, K; Bejder, L; Cerchio, S; Christiansen, F; Durban, J; Fearnbach, H; Fortune, S; Friedlaender, A; Kotik, C; Koski, WR; Miller, C; Rodríguez-González, FM; Segre, PS; Urban R, J; Vivier, F; Weir, CR; and MJ Moore. **Morphology of nares associated with stereo-olfaction in baleen whales.** *Biology Letters.*

2023 • Videsen, KA; Simon, M; Christiansen, F; Friedlaender, A; Goldbogen, J; Malte, H; Segre, P; Wang, T; Johnson, M; and PT Madsen. **Cheap gulp foraging of a giga-predator enables efficient exploitation of sparse prey.** *Science Advances.*

2023 • Segre, PS; Martin, J; Irschick, DJ; and JA Goldbogen. **A three-dimensional, dynamic blue whale model for research and scientific communication.** *Marine Mammal Science.*

2022 • Segre, PS; Gough, WT; Roualdes, EA; Cade, DE; Czapanskiy, MF; Fahlbusch, J; Kahane-Rapport, SR; Oestreich, WK; Bejder, L; Bierlich, KC et al. **Scaling of maneuvering performance in baleen whales: larger whales outperform expectations.** *Journal of Experimental Biology.*

2022 • Segre, PS; di Clemente, J; Kahane-Rapport, SR; Gough, WT; Meÿer, MA; Lombard, AT; Goldbogen, JA; and GS Penry. **High-speed chases along the seafloor put Bryde's whales at risk of entanglement.** *Conservation Science and Practice.*

2021 • Savoca, MS; Czapanskiy, MF; Kahane-Rapport, SR; Gough, WT; Fahlbusch, J; Bierlich, KC; Segre, PS; di Clemente, J; Penry, GS; Wiley, D et al. **Baleen whale prey consumption based on high resolution foraging measurements.** *Nature.*

2021 • Gough, WT; Smith, HJ; Savoca, MS; Czapanskiy, MF; Fish, FE; Potvin, J; Bierlich, KC; Cade, DE; di Clemente, J; Kennedy, J; Segre, PS et al. **Scaling of oscillatory kinematics and Froude efficiency in baleen whales.** *Journal of Experimental Biology.*

2021 • Segre, PS; Weir, CR; Stanworth, A; Cartwright, S; Friedlaender, AS; and JA Goldbogen. **Biomechanically distinct filter-feeding behaviors distinguish sei whales as a functional intermediate and ecologically flexible species.** *Journal of Experimental Biology.*

2021 • Czapanskiy, MF; Savoca, MS; Gough, WT; Segre, PS; Wisniewska, DM; Cade, DE; and JA Goldbogen. **Modelling short-term energetic costs of sonar disturbance to cetaceans using high-resolution foraging data.** *Journal of Applied Ecology.*

2020 • Caruso, F; Hickmot, L; Warren, JD; Segre, PS; Chiang, G; Bahamonde, P; Español-Jiménez, S; Li, S; and A. Bocconcelli. **Diel differences in blue whale (*Balaenoptera musculus*) dive behavior increases nighttime risk of ship strikes in northern Chilean Patagonia.** *Journal of Integrative Zoology.*

2020 • Dakin, R; Segre, PS; and DL Altshuler. **Individual variability and the biomechanics of maneuvering flight in hummingbirds.** *Journal of Experimental Biology.*

2020 • Kahane-Rapport, SR; Savoca, MS; Cade, DE; Segre, PS; Bierlich, KC; Calambokidis, JA; Dale, J; Friedlaender, AS; Johnston, DW; Werth, AJ; and JA Goldbogen. **Lunge filter feeding biomechanics constrain rorqual foraging ecology across scale.** *Journal of Experimental Biology.*

2020 • Segre, PS; Potvin, J; Cade, DE; Calambokidis, J; Di Clemente, J; Fish, FE; Friedlaender, AS; Gough, WT; Kahane-Rapport, SR; Oliveira, C et al. **Energetic and physical limitations on the breaching performance of large whales.** *eLife.*

2019 • Goldbogen, JA; Cade, DE; Wisniewska, DM; Potvin, J; Segre, PS; Savoca, MS; Hazen, EL; Czapanskiy, MF; Kahane-Rapport, SR; DeRuiter, SL et al. **Why whales are big but not bigger: physiological drivers and ecological limits in the age of ocean giants.** *Science.*

2019 • Friedlaender, AS; Bowers, MT; Cade, DE; Hazen, EL; Stimpert, AK; Allen, AN; Calambokidis, J; Fahlbusch, J; Segre, PS; Visser, F et al. **The advantages of diving deep: fin whales double their energy intake by targeting deep krill patches.** *Functional Ecology.*

2019 • Gough, WT; Segre, PS; Bierlich, KC; Cade, DE; Potvin, J; Fish, FE; Dale, J; di Clemente, J; Friedlaender, AS; Johnston, DW et al. **Scaling of swimming performance in the largest animals.** *Journal of Experimental Biology.*

2019 • Segre, PS and ED Taylor†. **Large ants do not carry their fair share: maximal load carrying performance of leaf-cutter ants (*Atta cephalotes*).** *Journal of Experimental Biology.*

2018 • Segre, PS; Cade, DE; Calambokidis, J; Fish, FE; Friedlaender, AS; Potvin, J; and JA Goldbogen. **Body flexibility enhances maneuverability in the world's largest predator.** *Integrative and Comparative Biology.*

2018 • Segre, PS and AI Banet. **The origin of avian flight: finding common ground.** *Biological Journal of the Linnean Society.*

2018 • Segre, PS; Bocconcelli, A; Hickmott, LS; Howes, G; Warren, JD; and G Chiang. **Offshore hummingbird sightings in Chilean Patagonia.** *Wilson Journal of Ornithology.*

2018 • Dakin, R*, Segre, PS*, Straw, AD; and DL Altshuler. **Morphology, muscle capacity, skill and maneuvering ability in hummingbirds.** *Science.*

2017 • Goller, B; Segre, PS; Middleton, KM; Dickinson, MH; and DL Altshuler. **Visual sensory signals dominate tactile cues during docked feeding in hummingbirds.** *Frontiers in Neuroscience.*

2017 • Saddler, MR; Bocconcelli, A; Hickmott, LS; Chiang, G; Landea-Briones, R; Bahamonde, PA; Howes, G; Segre, PS; and LS Sayigh. **Characterizing Chilean blue whale vocalizations with DTAGs: a test of using tag accelerometers for caller identification.** *Journal of Experimental Biology.*

2017 • Skandalis, DA; Segre, PS; Bahlman, JW; Groom, D; Welch Jr; KC; Witt, CC; McGuire, JA; Dudley, R; Lentink, D; and DL Altshuler. **Individual performance and the biomechanical origin of extreme wing allometry in hummingbirds.** *Nature Communications.*

2017 • Goldbogen, JA; Cade, DE; Boersma A; Calambokidis, J; Kahane-Rapport, S; Segre, PS; Stimpert, AS; and AS Friedlander. **Using digital tags with integrated video and inertial sensors to study moving morphology and associated behavior in large aquatic vertebrates.** *Anatomical Review.*

2017 • Segre, PS; Seakamela, SM; Meÿer, MA; Findlay, KP; and JA. Goldbogen. **Flapping flippers: evidence for a hydrodynamically active flipper-stroke in humpback whales.** *Current Biology.*

2017 • Goldbogen, JA; Cade, DE; Calambokidis, J; Friedlaender, AS; Potvin, J; Segre, PS; and AJ Werth. **How baleen whales feed: the biomechanics of engulfment and filtration.** *Annual Review of Marine Science.*

2016 • Segre, PS*, Dakin, R*, Read, TJG, and DL Altshuler. **Mechanical constraints on flight at high elevation decrease maneuvering performance.** *Current Biology.*

2016 • Segre, PS; Cade, DE; Fish, FE; Potvin, J; Allen, AN; Calambokidis, J; Friedlaender, AS; and JA Goldbogen. **Hydrodynamic properties of fin whale flippers predict maximum rolling performance.** *J. of Experimental Biology.*

2016 • Read, TJG*, Segre, PS*, Middleton, KM; and DL Altshuler. **Hummingbirds control turning velocity with body orientation and turning radius with asymmetrical wingbeat kinematics.** *J. Royal Soc. Interface.*

2015 • Segre, PS; Dakin, R; Zordan, VB; Dickinson, MH; Straw, AD; and DL Altshuler. **Burst muscle performance predicts the speed, acceleration, and turning performance of hummingbirds.** *eLife.*

2015 • Altshuler, DL; Bahlman, JW; Dakin, R; Gaede, AH; Goller, B; Lentink, D; Segre, PS; and DA Skandalis. **The biophysics of bird flight: functional relationships integrate aerodynamics, morphology, kinematics, muscles and sensors.** *Canadian Journal of Zoology.*

2012 • Pournazeri, S*, Segre, PS*, Princevac, M and DL Altshuler. **Hummingbirds generate bilateral vortex loops during hovering: evidence from flow visualization.** *Experiments in Fluids*.

2012 • Altshuler, DL; Quicazán-Rubio, EM; Segre, PS and KM Middleton. **Wingbeat kinematics and motor control of yaw turns in Anna's hummingbirds (*Calypte anna*).** *J. of Experimental Biology.*

2009 • Jackson, BE; Dial, KP; and PS Segre. **Precocial development of anti-predator locomotor performance in a ground-dwelling bird species (*Alectoris chukar*).** *Proc. Royal Soc. B.*

2008 • Dial, KP; Jackson, BE; and PS Segre. **A fundamental avian wing-stroke provides a new perspective on the evolution of flight**. *Nature*.

Technical Reports

2022 • Segre, PS and CR Weir. **Chapter 6: Suction cup tagging.** In *Conserving Falkland's whale populations: addressing data deficiencies for informed management* (ed. CR Weir).

Student Theses Supervised

2026 • Marissa Helgesen. Masters thesis. ***Anurans do not alter their calling behavior in response to traffic noise on the University of Wisconsin-Green Bay campus.*** *University of Wisconsin, Green Bay*

Grants Awarded

2026 • UWGB, NAS Holz Grant • Title: *Spatial movements of juvenile raptors in lower Green Bay* • Award: $3940

2026 • UWGB, Research Council Projects Grant • Title: *Niche partitioning and habitat use of baleen whales in the Galapagos Islands* • Award: $4400

2025 • NEW Bird Alliance • Title: *A new nest box for the UWGB peregrine falcons* • Award: $2500

2025 • Cofrin Center for Biodiversity, Freshwater Collaborative Internship for undergraduates • Title: *Comparative locomotor performance of anurans* • Award: $2150

2025 • UWGB, NAS Holz Grant • Title: *Cyclical habitat use by resident birds in Wequiock Creek Natural Area* • Award: $1735

2025 • UWGB, Research Council Projects Grant • Title: *Ecology, behavior, and conservation of deep diving cetaceans in the Ionian Sea* • Award: $2700

2024 • UWGB, NAS Holz Grant • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $2000

2024 • UWGB, Research Council Projects Grant • Title: *Foraging ecology of Southern right whales on a newly discovered feeding ground in western South Africa* • Award: $2900

2024 • University of Wisconsin Green Bay, Holtz Award • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $2000

2024 • University of Wisconsin Green Bay, Research Council Grant • Title: *Foraging ecology of Southern right whales on a newly discovered feeding ground in western South Africa* • Award: $2900

2022 • Chico State Affordable Learning Solutions program (CAL$) • Co-Principal Investigator • Grant to support adoption of open source and low cost educational materials for introductory biology • Award: $6,487

2022 • ArcticNet • Co-Principal Investigator • Title: *An ecosystem approach to quantifying behavioural and energetic impacts of anthropogenic disturbance to Arctic whales* • Award: $116,330

2020 • Dalio Foundation/OceanX • Principal Investigator • Title: *Foraging strategies and energetics of inshore South African Bryde's whales: implications for mitigation against lethal entanglement and prey availability* • Award: $62,580

2019 • Science Education Exploration Initiative, Sofar Ocean. • Principal Investigator • Title: *Assessing rorqual whale prey fields with a small underwater ROV* • Award: $1,695

2018 • PADI Foundation Grant for Marine Conservation • Collaborator: contributed to project conceptualization and proposal writing • Title: *Pursuing food and dealing with its cost: a fine-scale study on the foraging ecology of South African inshore Bryde's whales (Balaenoptera edeni) using camera-borne tags* • Award: $5,000

2017 • Society for Marine Mammalogy • Collaborator: contributed to project conceptualization and proposal writing • Title: *Pursuing food and dealing with its cost: a fine-scale study on the foraging ecology of South African inshore Bryde's whales (Balaenoptera edeni) using camera-borne tags* • Award: $2,000

2017 • NSF Division of Integrative Organismal Systems Grant • Collaborator: contributed to project conceptualization and proposal writing • Title: *Scaling of unsteady locomotor performance and maneuverability* • Award: $322,907

2012 • University of British Columbia • Hesse Research Award in Ornithology • Principal Investigator • Title: *The constraints of morphology and reserve power capacities on maneuvering performance in hummingbirds* • Award: $1,485

2009 • Valentine Eastern Sierra Reserve Graduate Student Grant • Principal Investigator • Title: *The effect of altitudinal gradients on hummingbird performance in two species of hummingbirds* • Award: $2,500

2009 • White Mountain Research Station Minigrant • Principal Investigator • Title: *The effects of elevation on maneuverability in two species of hummingbirds* • Award: $2,000

2009 • NSF Division of Integrative Organismal Systems Grant • Collaborator: contributed to project conceptualization and proposal writing • Title: *Computational analysis of maneuvering flight* • Award: $163,173

Grants Awarded to Students

2026 • UWGB, NAS Research Assistantship • to Edwin Wilke, M.S. student

2026 • UWGB, Cofrin Center for Biodiversity Student Grant • to Janelle Lopez, Masters student • Title: *Restoration in Flight: Peregrine Falcons and Public Engagement at the University of Wisconsin- Green Bay* • Award: $3500

2026 • UWGB, Cofrin Center for Biodiversity Student Grant • to Roberta Fenton, Masters student • Title: *Owl occupancy at Toft Point in Door County, Wisconsin* • Award: $3065

2026 • UWGB, Cofrin Center for Biodiversity Student Grant • to Abbie Adams, Undergraduate student • Title: *The effects of outdoor recreation on the activity patterns of medium-sized mammals in Northeast Wisconsin: Year 2 of the UWGB Snapshot USA survey* • Award: $3060

2026 • UWGB, Cofrin Center for Biodiversity Student Grant • to Edwin Wilke, M.S. student • Title: *Habitat use and diet of songbirds at Wequiock Creek and Point au Sable natural areas* • Award: $3000

2026 • UWGB, NAS Heirloom Grant • to Daniel Erickson, M.S. student • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $500

2025 • UWGB, NAS Heirloom Grant • to Daniel Erickson, M.S. student • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $1500

2025 • UWGB, NAS Research Assistantship • to Daniel Erickson, M.S. student

2025 • UWGB, Cofrin Center for Biodiversity Student Grant • to Megan Oliva, undergraduate student • Title: *Mammal Monitoring in the Point au Sable and Wequiock Creek Natural Areas* • Award: $2500

2025 • UWGB, Cofrin Center for Biodiversity Student Grant • to Edwin Wilke, M.S. student • Title: *Habitat use and diet of songbirds at Wequiock Creek and Point au Sable natural areas* • Award: $3000

2025 • UWGB, Cofrin Center for Biodiversity Student Grant • to Gweni Malokofsky and Ethan Fenlon, undergraduate students co-advised with Erin Giese • Title: *Breeding Birds and Small Mammals of Kingfisher Farm Natural Area* • Award: $4000

2024 • UWGB, NAS Heirloom Grant • to Daniel Erickson, M.S. student • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $2000

2024 • University of Wisconsin Green Bay, Natural and Applied Sciences Heirloom Grant • to Marissa Helgesen, M.S. student • Title: *Anuran calling behavior in response to traffic noise* • Award: $1450

2024 • University of Wisconsin Green Bay, Cofrin Center for Biodiversity Student Grant • to Marissa Helgesen, M.S. student • Title: *Anuran calling behavior in response to traffic noise* • Award: $2500

2024 • University of Wisconsin Green Bay, Graduate Student Research Grant • to Daniel Erickson, M.S. student • Title: *Population connectivity and the spread of communicable disease in sharp-shinned hawks* • Award: $1000

Teaching Experience

2023 - Present • University of Wisconsin, Green Bay

BIO 342 • Ornithology

BIO 343 • Mammalogy

BIO 340 • Comparative Anatomy of the Vertebrates

BCM 700 • Conservation Ecology

BCM 705 • Conservation Research and Management

BCM 725 • Evolution, Biodiversity, and Conservation • co-instructor

2020 - 2023 • California State University, Chico

Advanced Zoology • Lab

Mammalogy

Ornithology

Introduction to living systems

2018 • Organization for Tropical Studies

Research Experience for U.S. Underrepresented Minority Students in Costa Rica

2017 - 2018 • Stanford University

Bio-logging and bio-telemetry • co-instructor

Marine megafauna • co-instructor

2004 - 2015 • Teaching Assistantships

Ecology lab • University of British Columbia

Introductory biology lab • University of California, Riverside

Comparative vertebrate anatomy • University of California, Riverside

Introductory biology lab • University of Montana

Ecology lab • University of Montana

Parasitology lab • University of Montana

Student Supervision - Current

2024 - Present • Daniel Erickson. Masters student. **Population connectivity and the spread of communicable disease in sharp-shinned hawks.** University of Wisconsin, Green Bay

2024 - Present • Megan Oliva. Undergraduate student. **Mapping noise pollution on the UWGB campus; Mammal monitoring in the Point au Sable and Wequiock Creek Natural Areas.** University of Wisconsin, Green Bay

2025 - Present • Edwin Wilke. Masters student. **Habitat use and diet of songbirds at Wequiock Creek and Point au Sable Natural Areas.** University of Wisconsin, Green Bay.

2025 - Present • Abbie Adams, Undergraduate student. **The effects of outdoor recreation on the activity patterns of medium-sized mammals in Northeast Wisconsin: Year 2 of the UWGB Snapshot USA survey.** University of Wisconsin, Green Bay

2026 - Present • Janelle Lopez, Masters student. **Causes of juvenile mortality in red-shouldered hawks (*Buteo lineatus*) of the upper Midwest.** University of Wisconsin, Green Bay.

2026 - Present • Roberta Fenton, Masters student. **Owl occupancy at Toft Point in Door County, Wisconsin.** University of Wisconsin, Green Bay.

Student Supervision - Past

2026 • Gweni Malokofsky. Undergraduate student. **Small mammals of Kingfisher Farm Natural Area.** University of Wisconsin, Green Bay

2026 • Marissa Helgesen. Masters student. **Anurans do not alter their calling behavior in response to traffic noise on the University of Wisconsin-Green Bay campus.** University of Wisconsin, Green Bay

2025 • Jared Ganeles. Masters student. **Impacts of habitat restoration on migratory and resident bird populations.** University of Wisconsin, Green Bay

2025 • Olivia Castruita. Undergraduate student. **Mapping noise pollution on the UWGB campus.** University of Wisconsin, Green Bay

2023 • Tiffany Paalman. Undergraduate student. **Environmental determinants of whale barnacle behavior.** University of Wisconsin, Green Bay

2023 • Charlie Giannini. Undergraduate student. **Body condition of saw-whet owls at the Big Chico Creek Ecological Reserve.** California State University, Chico

2023 • Janae Dugas. Undergraduate student. **Whale lice behavior.** California State University, Chico

2023 • Isabelle Fabbro. Undergraduate student. **Jaw muscle allocation in California ground squirrels.** California State University, Chico

2022 • Masha Elson. Undergraduate student. **Environmental determinants of whale barnacle behavior.** California State University, Chico

2018 • Ebony Taylor. NSF-LSAMP REU student. **Maximal load carrying performance of Leaf-Cutter Ants (*Atta cephalotes*).** Organization for Tropical Studies, Costa Rica

2018 • Gabriel Martinez. NSF-LSAMP REU student. **Aerial maneuverability of the Social flycatcher during aerial sallies.** Organization for Tropical Studies, Costa Rica

2015 • Tyson Read. M.Sc. **Hummingbirds use banking to achieve faster turns and asymmetrical wingstrokes to achieve tighter turns.** University of British Columbia

2012 • James Whale. B.Sc. Honors Thesis. **Kinematics of bar-headed geese in hypoxia.** University of British Columbia

2010 • Lev Darkhovsky. CNAS Dean's Summer Research Fellowship. **Morphology and maneuverability in tropical hummingbirds.** University of California, Riverside

2010 • Adam Behroozian. B.Sc. Honors Project. **Competitive interactions between male Anna's hummingbirds (*Calypte anna*): understanding the roles of morphology, power margins, and behavior in predicting dominance.** University of California, Riverside

Student Advisory Committees

2026 • Morley Remitz. M.Sc. **Habitat selection of long-jawed orb-weaving spiders (*Tetragnathidae: Tetragnatha spp.*) across riparian-to-upland gradients at Wequiock Creek, Brown County, Wisconsin.** University of Wisconsin - Green Bay.

2025 • Kadisyn Ceplina. M.Sc. **Comparative assessment of fish and macroinvertebrate biomonitoring sampling techniques for streams in northeastern Wisconsin.** University of Wisconsin - Green Bay.

2023 • Jay Kirkham. Ph.D. **North Atlantic right whale foraging ecology.** Dalhousie University.

2023 • Manon den Haan. Ph.D. **Bowhead whale foraging ecology.** Dalhousie University.

2022 • Nicholas Balfour. M.Sc. **High incubation temperature and thermal stress increase the aerobic performance of juvenile fall-run Chinook salmon (*Oncorhynchus tshawytscha*).** California State University, Chico.

Field Expeditions

Jan 2026, Jul 2026 • Foraging ecology of baleen whales in the Galapagos Islands. Galapagos, Ecuador.

Aug 2025 • Deep diving whales in the Mediterranean. Sicily, Italy

Aug 2024, Aug 2023 • Bowhead whale foraging ecology. Baffin Island, Canada

Jan 2023, Jan 2025 • Southern right whale foraging ecology. South Africa

Mar 2019 • Sei whale foraging ecology and biomechanics. Falkland Islands

Jun 2018 • NSF LSAMP REU program. La Selva Biological Station, Costa Rica

Apr 2018 • Bryde's whale foraging ecology and biomechanics. Plettenberg Bay, South Africa

Sep 2017 • Subarctic whale ecology and biomechanics. Greenland Institute of Natural Resources, Greenland

Aug 2017, Aug 2018 • Rorqual whale maneuverability. Channel Islands and Monterey Bay, California

Feb 2016, Feb 2017 • Patagonia blue whale ecology and biomechanics. Fundacion MERI, Chile

Sep 2016 • Arctic blue whale ecology and biomechanics. Norwegian Polar Institute, Svalbard

Jun 2013 • High elevation Andean hummingbird maneuverability. Hacienda Guaytara, Ecuador

Feb 2012 • High and low elevation tropical hummingbird maneuverability. La Selva Biological Station and Cerro De La Muerte, Costa Rica

Jul 2010 • Amazonian hummingbird maneuverability. Los Amigos Field Station, Peru

May 2009 • High elevation hummingbird maneuverability. Valentine Eastern Sierra Reserve and White Mountain Research Station, California

Awards

2015 • Pacific Ecology and Evolution Conference • Best short talk award, runner up

2013 • Cooper Ornithological Society • Katma Award for unconventional and innovative ideas in ornithology

2011, 2012 • University of British Columbia • Department of Zoology Graduate Fellowship

2008 • University of California, Riverside • Chancellor Distinguished Fellowship

Conferences and Workshops

2020 • Society Integrative and Comparative Biology: talk presented

2018 • NSF-LSAMP Diversity and mentoring workshop, Organization for Tropical Studies

2018 • Teaching workshop for Postdoctoral Scholars, Stanford University

2018 • Society Integrative and Comparative Biology: talk presented

2017 • Inertial Sensing Workshop, Kerteminde Denmark

2017 • Bio-logging Symposium: poster presented

2017 • Center for Integrated Research Teaching and Learning online course: Introduction to evidence-based undergraduate STEM teaching

2016 • Society Integrative and Comparative Biology: talk presented

2015 • Pacific Ecology and Evolution Conference: talk presented

2012 • Society for Integrative and Comparative Biology: symposium talk co-author

2008 • Desert Tortoise Research Council

2006 • Society Integrative and Comparative Biology: poster presented

2005 • American Ornithologists Union

Invited talks

Jan 2025 • Universidad San Francisco de Quito, Galapagos, invited speaker

Mar 2019 • Falkland Conservation Seminar Series, invited speaker

Sep 2018 • Monterey Birding Festival, invited speaker

Mar 2016 • Chico State, biology seminar series

Oct 2006 • Illinois Institute of Technology, biology colloquium series

Work Experience

Feb 2008 • Field Crew Leader, Great Basin Institute: Mojave desert tortoise line distance survey, Nevada

Aug 2007 • Field Technician, Australian Wildlife Conservancy: Purple crowned fairy wren distribution and genetic survey, Mornington Wildlife Research Station, Australia

Mar 2007 • Field Technician, Montana Cooperative Wildlife Research Unit and the University of Montana: long term research of life history variation in tropical birds, Yacumbu, Venezuela

Apr 2004 • Field Technician, United States Geological Survey and University of California at Davis: spotted owl survey in Yosemite National Park

Jan 2004 • Field Intern, Minnesota Department of Natural Resources: Winter population ecology of white tailed deer and gray wolves

Aug 2003 • Field Intern, United States Geological Survey, Hawaii: Biocomplexity of avian disease

May 2003 • Field and Lab Technician, University of Illinois- Smithsonian Institute for Tropical Studies- Barro Colorado Island, Panama: tropical forest soil dynamics

May 2002 • Undergraduate Research Assistant, Illinois Institute of Technology – Biophysics Collaborative Access Team, Argonne National Laboratory: drosophila flight muscle structure

Professional Qualifications

2025 - Present • Federal Bird Banding subpermittee • United States Geological Survey

2017 - Present • Remote pilot certification • Federal Aviation Administration

2009 - 2017 • Federal Bird Banding subpermittee • United States Geological Survey

2006 • Desert Tortoise handling permittee • United States Fish and Wildlife Service

Popular Press

2025 • Huge if True: *Could ants carry a person? We asked experts!*; Cleo Abram

2025 • *The Right Whale;* by Spindrift Images

2025 • Science News Explores: *To clear loops, Sonic the Hedgehog must hit the right speed*; Aaron Tremper

2024 • New York Times: *On the wings of war;* Jim Robbins

2023 • Science News Explores: *Meet robots on a mission to help birds;* Aaron Tremper

2023 • El Pais: *Este es el secreto de los colibríes para encontrar comida y refugio;* Veronica M. Garrido

2022 • Inside JEB: *Gargantuan whales more manoeuvrable than they ought to be;* Kathryn Knight

2021 • Nature News and Views: *A whale of an appetite revealed by analysis of prey consumption;* Victor Smetacek

2021 • Inside JEB: *Versatile sei whales both lunge and skim to feed;* Kathryn Knight

2020 • eLife Insights: *Whale breaching says it loud and clear;* Alexander J. Werth and Charles L. Lemon

2019 • Science Perspective: *The biology of big;* Terrie M. Williams

2019 • Audubon Magazine: *Hummingbird hackers;* Jessica Leber

2018 • Science Perspective: *How hummingbirds stay nimble on the wing;* Peter C. Wainwright

2018 • Science: *High-speed cameras reveal how hummingbirds can turn on a dime;* Elizabeth Pennisi

2018 • Seeker: *The secrets of hummingbird agility revealed*; Jen Viegas

2017 • National Geographic: *Rare video captures never-before-seen whale behavior;* Carrie Arnold

2017 • Scientific American Podcast: *Humpback whale flippers do more than maneuver;* Emily Schwing

2017 • Science News For Students: *Humpbacks flap their flippers like underwater birds;* Sharon Oosthoek

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Journal of the Royal Society, Interface	eLife
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